

## EXERCISE

CONSIDER THE FOLLOWING STOCHASTIC PROCESS:

$$y(t) = 2 \cdot e(t-1) + \frac{2}{3} e(t-2) \quad e \sim \mathcal{WN}(0, 1)$$

⇒ CLASSIFY THE PROCESS

MA(2) ⇒ STATIONARY "BY DEFINITION"

$$W(z) = 2z^{-1} + \frac{2}{3}z^{-2} = \frac{2z + \frac{2}{3}}{z^2}$$

⇒ WRITE THE PROCESS IN CANONICAL FORM

1.  $C(z), A(z)$  HAVE THE SAME ORDER ✗  $W(z) = \frac{C(z)}{A(z)}$
2.  $C(z), A(z)$  MONIC ✗
3.  $C(z), A(z)$  COPRIME ✓
4.  $C(z), A(z)$  ROOTS AND ZEROS UNITARY CIRCLE ✓  
    / ZERO:  $z = -1/3$   
    \ POLES:  $z_{1,2} = 0$

LET'S ADDRESS POINT 1, 2:

$$y(t) = [2z^{-1} + \frac{2}{3}z^{-2}] \cdot e(t)$$

$$= [2 + \frac{2}{3}z^{-1}] \cdot z^{-1} \cdot e(t)$$

$$= [2 + \frac{2}{3}z^{-1}] \cdot \tilde{e}(t)$$

$$= \frac{1}{1 + \frac{1}{3}z^{-1}} \cdot 2 \cdot \tilde{e}(t)$$

$$y(t) = [1 + \frac{1}{3}z^{-1}] \cdot \bar{e}(t)$$

$$\tilde{e}(t) \sim \mathcal{WN}(0, 1)$$

$e(t)$  AND  $\tilde{e}(t)$  ARE STOCHASTICALLY SPEAKING THE SAME PROCESS

$$\bar{e}(t) \sim \mathcal{WN}(0, 1 \cdot 2^2) \quad \sim 4$$

( $\bar{W}(z)$  IN CANONICAL FORM!

$$\left. \begin{aligned} y(t) &= W(z) \cdot e(t) \\ y(t) &= \bar{W}(z) \cdot \bar{e}(t) \end{aligned} \right\} \text{EQUIVALENT PROCESSES FROM A STOCHASTIC POINT OF VIEW}$$

$\bar{W}(z)$  IS MA(1) ⇒ EASIER!

## ⇒ DESCRIBES $y(t)$ IN TERMS OF ITS STOCHASTIC PROPERTIES

### • MEAN VALUE

$$\begin{aligned}\mu_y &= \tilde{W}(z=1) \cdot \mu_{\tilde{e}} = 0 \\ &= W(z=1) \cdot \mu_e = 0\end{aligned}$$

### • COVARIANCE FUNCTION

$$\gamma(0) = \underbrace{\left[1 + \frac{1}{3}\right]}_{c_0 + c_1} \cdot 4 \stackrel{\sim}{=} \frac{40}{3} \quad \text{VARIANCE OF } y(t)$$

$$\gamma(1) = c_0 \cdot c_1 \cdot 4 = \frac{4}{3}$$

$$\gamma(z \geq 2) = 0$$

## ⇒ COMPUTE THE PSD $\left[\Gamma_y(\omega)\right]$ AND DRAW ITS DIAGRAM

1. DEFINITION  $\Gamma_y(\omega) = \sum_{-\infty}^{+\infty} \gamma(z) \cdot e^{-j\omega z}$  \* SUITABLE ONLY FOR MA PROCESSES

$$\begin{aligned}\Gamma_y(\omega) &= \gamma(-1) \cdot e^{j\omega} + \gamma(0) e^{-0} + \gamma(1) e^{-j\omega} \\ &= \frac{40}{3} + \frac{4}{3} \cdot [e^{j\omega} + e^{-j\omega}] \quad \text{EULER FORMULA}\end{aligned}$$

$$\Gamma_y(\omega) = \frac{40}{3} + \frac{8}{3} \cdot \cos \omega$$

$$(a + bj)(a - bj) = a^2 + b^2$$

### 2. TRANSFER FUNCTION

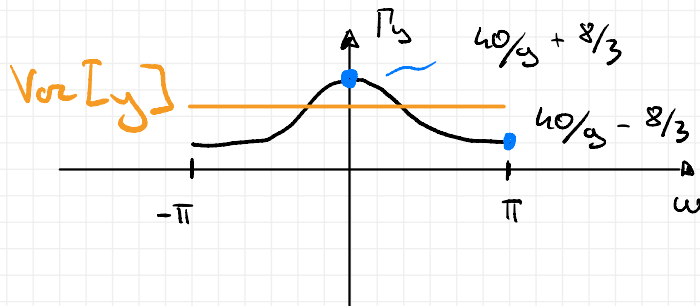
$$\Gamma_y(\omega) = \left| \bar{W}(z=e^{j\omega}) \right|^2 \cdot \Gamma_{\tilde{e}}(\omega)$$

$$\begin{aligned}&= \underbrace{\left[1 + \frac{1}{3} e^{-j\omega}\right]}_{\bar{W}(e^{j\omega})} \underbrace{\left[1 + \frac{1}{3} e^{j\omega}\right]}_{\bar{W}(e^{-j\omega})} \cdot 4 \quad \text{Var}[\tilde{e}(t)] \\ &= \left[1 + \frac{1}{3} + \frac{1}{3} (e^{j\omega} + e^{-j\omega})\right] \cdot 4\end{aligned}$$

$$= \left[ \frac{40}{9} + \frac{2}{3} \cdot \cos \omega \right] \cdot 4$$

$$\Gamma_y = \frac{40}{9} + \frac{8}{3} \cdot \cos \omega$$

LET'S DRAW THE DIAGRAM:



SIMPLE  
COSINUSOIDAL  
FUNCTION

★ COMPUTES THE PROCESS VARIANCE, STARTING FROM THE PSD

$$\gamma(\tau) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma(\omega) \cdot e^{j\omega\tau} d\omega \quad \text{INVERSE FUNCTION}$$

$$\downarrow$$

$$\text{Var}[y(t)] = \gamma(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Gamma(\omega) d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[ \frac{40}{9} + \frac{8}{3} \cos \omega \right] d\omega$$

$$= \frac{2\pi}{2\pi} \cdot \frac{40}{9} + \frac{1}{2\pi} \cdot 0 \quad \begin{array}{l} \text{INTEGRAL OF } \cos \omega \text{ OVER} \\ \text{AN ENTIRE PERIOD} \end{array}$$

$$= \frac{40}{9}$$

## EXERCISE

CONSIDER THE FOLLOWING STOCHASTIC PROCESS:

$$y(t) = \frac{1}{6} y(t-1) + \frac{1}{6} y(t-2) + \eta(t) + \frac{1}{2} \eta(t-1) \quad \eta \sim \text{WN}(0, 2)$$

⇒ CLASSIFY THE PROCESS

• ARMA(2, 1)

•  $\eta(t)$  IS A S.S.P

$$\bullet W(z) = \frac{1 + \frac{1}{2} z^{-1}}{1 - \frac{1}{6} z^{-1} - \frac{1}{6} z^{-2}} = \frac{(z + \frac{1}{2}) z}{z^2 - \frac{1}{6} z - \frac{1}{6}}$$

$$\text{POLES: } z_{1,2} = \frac{\frac{1}{6} \pm \sqrt{\frac{1}{36} + \frac{4}{6}}}{2} \quad \begin{cases} z = \frac{1}{2} \\ z = -\frac{1}{3} \end{cases}$$

⇒ BOTH POLES ARE INSIDE THE UNITARY CIRCLE

⇓  
 $W(z)$  IS A.S.

⇓  
 $y(t)$  IS STATIONARY

⇒ PROVIDE THE STOCHASTIC DESCRIPTION OF THE PROCESS

• MEAN VALUES

$$\mu_y = 0, \text{ SINCE } \mu_\eta = 0$$

• COVARIANCE FUNCTION

$$\gamma(z) = \mathbb{E}[y(t) \cdot y(t-z)]$$

$$= \mathbb{E}\left[\left(\frac{1}{6} y(t-1) + \frac{1}{6} y(t-2) + \eta(t) + \frac{1}{2} \eta(t-1)\right) \cdot y(t-z)\right]$$

$$= \frac{1}{6} \mathbb{E}[y(t-1) \cdot y(t-z)] + \frac{1}{6} \mathbb{E}[y(t-2) \cdot y(t-z)] + \mathbb{E}[\eta(t) \cdot y(t-z)] + \frac{1}{2} \mathbb{E}[\eta(t-1) \cdot y(t-z)]$$

$$\gamma(z) = \frac{1}{6} \cdot \gamma(z-1) + \frac{1}{6} \gamma(z-2) + \square + \frac{1}{2} \square$$



$$\mathbb{E}[v(t) \cdot y(t-z)] \begin{cases} z=0: \lambda_v^2 \\ z>0: 0 \end{cases}$$

$$\bullet z=0: \mathbb{E}[v(t) \cdot y(t)] = \mathbb{E}\left[v(t) \cdot \left(\frac{1}{6}y(t-1) + \frac{1}{6}y(t-2) + v(t) + \frac{1}{2}v(t-1)\right)\right]$$

$$= \lambda_v^2 + 0$$

$$\bullet z>0: 0$$

$$\mathbb{E}[v(t-1) \cdot y(t-z)]$$

$$\bullet z=0: \mathbb{E}[v(t-1) \cdot y(t)] = \mathbb{E}\left[v(t-1) \left(\frac{1}{6}y(t-1) + \frac{1}{6}y(t-2) + v(t) + \frac{1}{2}v(t-1)\right)\right]$$

$$= \frac{1}{2} \cdot \lambda_v^2 + \frac{1}{6} \cdot \lambda_v^2 \sim \mathbb{E}[v(t-1) \cdot y(t-1)] = \mathbb{E}[v(t) \cdot y(t)]$$

$$= \frac{4}{6} \lambda_v^2$$

$$\bullet z=1: \mathbb{E}[v(t-1) \cdot y(t-1)] = \lambda_v^2$$

$$\bullet z \geq 2: \mathbb{E}[v(t-1) \cdot y(t-z)] = 0$$

In summary, we get:

$$\bullet \gamma(0) = \frac{1}{6} \gamma(1) + \frac{1}{6} \gamma(2) + \lambda^2 + \frac{4}{12} \lambda^2 = \frac{1}{6} \gamma(1) + \frac{1}{6} \gamma(2) + \frac{4}{3} \lambda^2$$

$$\bullet \gamma(1) = \frac{1}{6} \gamma(0) + \frac{1}{6} \gamma(1) + 0 + \frac{1}{2} \lambda^2$$

$$\bullet \gamma(z \geq 2) = \frac{1}{6} \gamma(z-1) + \frac{1}{6} \gamma(z-2)$$

RECURSIVE EQUATION, TYPICAL OF AR PART OF THE PROCESS

LET'S COMPUTE THE "INITIAL CONDITIONS":

$$\bullet \gamma(0) = \frac{1}{6} \gamma(1) + \frac{1}{6} \gamma(2) + \frac{4}{3} \lambda^2$$

$$\bullet \gamma(1) = \frac{1}{6} \gamma(0) + \frac{1}{6} \gamma(1) + \frac{1}{2} \lambda^2$$

$$\bullet \gamma(2) = \frac{1}{6} \gamma(1) + \frac{1}{6} \gamma(0)$$

LINEAR SYSTEM OF EQUATIONS

3 UNKNOWN, 3 EQ.

$\Downarrow$

"INITIAL CONDITIONS"

$$\gamma(0) = \frac{87}{28}$$

$$\gamma(1) = \frac{255}{140}$$

$$\gamma(2) = \frac{1}{6} \gamma(1) + \frac{1}{6} \gamma(0)$$

RECURSIVE EQUATION

TYPICAL STRUCTURE OF SP WITH "AR" PART:

- "INITIAL CONDITIONS"
- RECURSIVE EQUATION

⇒ COMPUTE AND DRAW THE PSD OF  $y(t)$

$$\begin{aligned}
 P_y(\omega) &= \frac{1 + 1/2 e^{-j\omega}}{1 - 1/6 e^{-j\omega} - 1/6 e^{-2j\omega}} \cdot \frac{1 + 1/2 e^{j\omega}}{1 - 1/6 e^{j\omega} - 1/6 e^{2j\omega}} \cdot 2 \\
 &= \frac{1 + 1/4 + \cos \omega \sim \frac{1}{2} \cdot [e^{j\omega} + e^{-j\omega}]}{5/2 + 2 \cos \omega} \cdot 2 \\
 &= \frac{12/18 - 5/18 \cos \omega - 1/3 \cos(2\omega)}{12/18 - 5/18 \cos \omega - 1/3 \cos(2\omega)}
 \end{aligned}$$

TO DRAW THE PSD WE EVALUATE  $P(\omega)$  IN "SOME" POINTS

$$P(\omega=0) = \frac{81}{8} \sim 10$$

$$P(\omega=\pi/2) = \frac{18}{10} = 1,8$$

$$P(\omega=\pi) = 0,5$$

